Statistical Learning
Second Homework: Supervised Learning
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1. Objectives

For this homework we decided to choose the R-library open dataset NHANES (National Health and Nutrition Examination Survey). It is a dataset that shows the health data of survey takers in the year interval 2009-2010 and 2011-2012. This dataset has 10000 observations and 76 variables. We want to explore our data using different supervised learning tools (Bayes Classifiers, Logistic regression, machine learning tools, feature importance). We aim to see how good of a model we can create, which will be able to classify if a patient has diabetes or not depending on his socioeconomic and medical data.

2. Data processing and visualization

a. Variable selection

Looking at the summary of the data, we can see that there are columns that have a lot of NAs. For our purposes, we can remove those features. We decided to keep both numerical and categorical values.

b. Dealing with NA values

Find which rows have the most NA values and remove the rows that have a lot of missing data. Depending on the variable we will either remove all rows with NA values(Diabetes), use the mode to fill the NAs(HHIncome), fill the missing values(categorical) according to the ages of the people (Marital status, Physical activity), or fill the missing values for the numerical variables with the average of every variable for the appropriate age range (weight, height, BPSysAve, etc).

c. Feature engineering

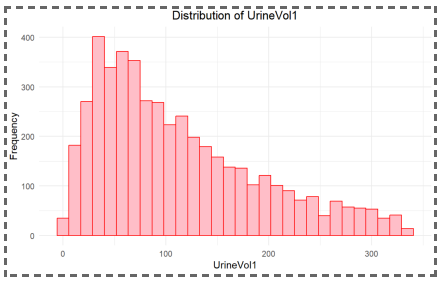
- SleepAdjusted shows if based on age, a person is sleeping enough. Remove SleepHrsYears.
- BMI – use the values from BMICatUnder20yrs which are not NA and for NA classify based on the BMI whether a person is Obese, OverWeight, NormWeight or UnderWeight.

d. Dealing with outliers

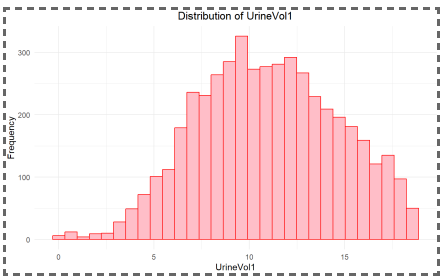
We have many outliers in one of the columns(UrineFlow1) so we have decided to remove this feature. After removing that variable, we can now remove the outliers of the remaining variable.

e. Distribution and Transformation

Create histograms to show the distribution of each variable, allowing us to see which variables are normally distributed, which are skewed, which are multimodal(have peaks in the histogram), or which have low variance. Overall, most variables are normally distributed, except UrineVol1, which exhibits heavy right skewness. We can transform these variables to get them to be more normally distributed.



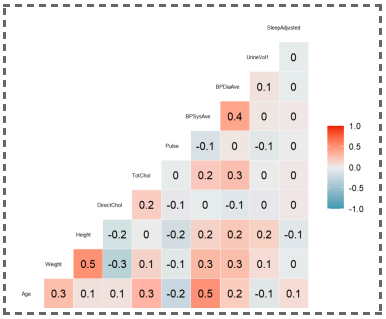
Before



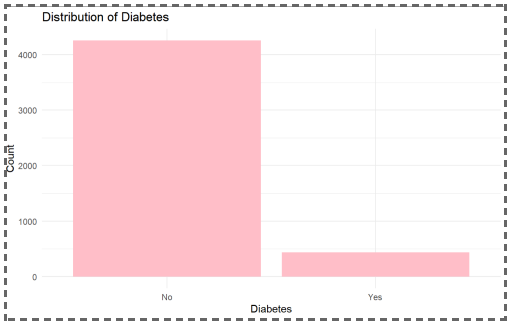
After

f. Visualization

i. Correlations between predictors. Moderate correlation



ii. Distribution of Diabetes



This means that our dataset is unbalanced so we will need to be more careful with interpreting the results of our models and make sure that our model is not choosing wrongly just to get a better overall accuracy score. We will focus on specificity as in our case that specifies how well our models are predicting the people that have diabetes.

iii. Distribution of variables based on diabetes

1. Creating bar plots for categorical variables, we find that gender doesn't really have an effect, black people are the most likely to have diabetes (but the significance of race isn't that high), obese and overweight people are more likely to be diabetic, 25% of widowed people and roughly 20% of divorced and separated people have diabetes, income doesn't favor one group, but people with lower income are more likely to have it, and not physically active people are also more likely to be diabetic.
2. Creating density plots for numerical data shows that older people, people that weigh more or people that have higher BPSysAve are more likely to be diabetic. Otherwise, the other variables don't really show a particular impact on the diagnosis

g. Normalization - we have normalized every numerical variable that we have

3. Bayes Classifiers

a. Linear discriminant analysis

The coefficients of the linear discriminants show that the most important predictors are Age, Race, Marital Status, while moderate predictors are Weight, Height, HHIncome, BPSysAve, BPDiaAve. The rest don't contribute as much.

```
          Reference
Prediction   No  Yes
       No  2463  198
       Yes   88   65
 Accuracy
0.8983653
Specificity
  0.2471483
Balanced Accuracy
         0.606326
```

The model:

This shows us that the model is only able to correctly identify 24.7% of the people that have diabetes. This is not very good as False negative results can cause a lot of harm to a person.

b. Quadratic discriminant analysis

The most important variables are Age, BMIObese, Race variables, and Income.

```
          Reference
Prediction   No  Yes
       No  2194  129
       Yes  357  134
 Accuracy
0.8272921
Specificity
  0.5095057
Balanced Accuracy
        0.6847803
```

The accuracy is not bad. The model correctly identifies 50.9% of individuals with diabetes, which is better than for LDA. While LDA has overall better accuracy, QDA has better balanced accuracy and specificity, which indicates that this model is less biased than the previous one.

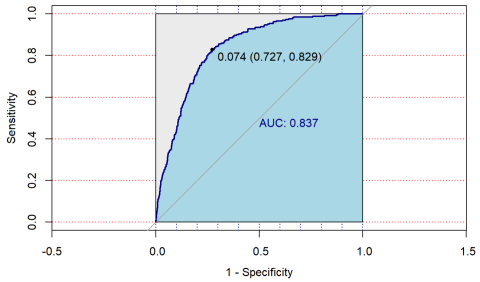
4. Logistic regression and penalized logistic regression

a. <u>Logistic regression</u>

- Create a logistic regression model and use the summary to find which variables are significant with p-value<0.05. These variables are Age, Weight, Height, TotChol, Pulse, BPSysAve, and SleepAdjusted.

- Creating a new model with these variables as the formula and creating the predictions with probability threshold of 0.5, we find that the model has specificity of only 11.03%, which is very low.

- Using the ROC curve, we are able to obtain a better probability threshold of 0.074, which gives us a model that can correctly identify 82.89% of the people that have diabetes. This is a very good model as it means that there are very few False Negatives, which is exactly what we are trying to obtain.



b. <u>Penalized logistic regression</u>

- Using loops, we are able to find the values for alpha and lambda, which will give us the best specificity and balanced accuracy. In our case these values are:

	alpha <dbl>	lambda <dbl>	balanced_accuracy <dbl>	specificity <dbl>	threshold <dbl>
Balanced Accuracy347	0.8	0.007943282	0.7897514	0.8821293	0.07610868

This is a very high specificity and good balanced accuracy, which is exactly what we were aiming for. While the balanced accuracy shows that the sensitivity is not as good, we are more focused on getting the "Yes" cases correct as False Positives cause way less harm to a person's health. Hence, this is an even better model than the simple logistic regression model.

5. Heath impact based on penalized logistic regression

- True Positive: Early diagnosis can be very beneficial to a person's health. We will assign the health impact of a true positive to be +10.

- False Positive: Unnecessary treatments and stressful situation. We will assign the health impact of a false positive to be -1.

- False Negative: Misdiagnosis could cause severe complications and increased chance of death. The assigned health impact will be -15.

- True Negative: No harm or benefit. The health impact will be 0.

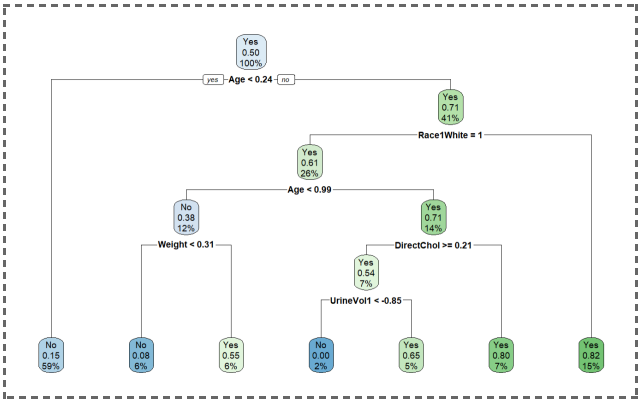
Using the previous penalized logistic regression model, we get an average of 0.385, which shows that we have a positive outcome overall and the guessed true positive outweighs the false negative and false positives.

6. **Decision Tree**

Using loops we are able to tune the hyper parameters minsplit, maxdepth, and cp to get the best specificity and balanced accuracy. We get the following results:

	minsplit <dbl>	maxdepth <dbl>	cp <dbl>	specificity <dbl>	balanced_accuracy <dbl>
Specificity42	120	6	0.001	0.8212928	0.7687413

This model is rather good based on the high specificity. The tree created with this parameters is:



This tree shows a local view of how the most important variables are utilized at the decision-making steps. Here, the most important features are:

- Age
- Race1
- Weight
- DirectChol
- UrineVol1

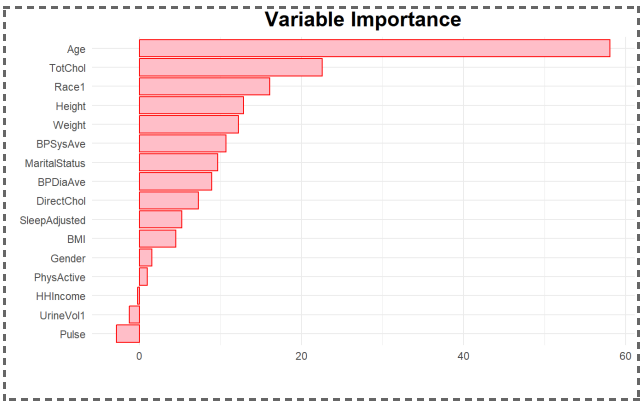
Using varImp(), we can see the overall importance of all the variables and which variables influence the model the most. Age is the most important, followed by BPSysAve. BMIObese, Weight, and MaritalStatusNeverMarried have similar significance and TotChol and DirectChol do as well. The rest of the variables have much lower significance.

7. **Random Forest**

Using loops we are able to tune the hyper parameters ntree and mtry to get the best specificity and balanced accuracy. We get the following results:

	mtry <dbl>	ntree <dbl>	accuracy <dbl>	specificity <dbl>	balanced_accuracy <dbl>
Accuracy58	16	1000	0.9051173	0.1634981	0.572537

This is not a good model as it is only able to correctly identify 16.35% of the diabetic patients. The variable importance of this model is:



The most important variable is Age, followed by TotChol, Race, Height, Weight, and BPSysAve.

8. **Gradient Boosting Model (GBM)**

This model was trained using the 'training_data' dataset and the distribution is set to Bernoulli because the target variable is binary. Then, predictions are made on 'testing_data' and the confusion matrix is computed. Looking at the metrics obtained (specificity = 0.74, balanced accuracy = 0.76) show that this model may not be the best, but has good results.

9. **Risk learning**

Using GBM we calculated the risk for each patient based on Diabetes probability, assuming a higher probability corresponds to a higher risk. We computed the percentage of patients with high risk, which is around 26%, and then looked for the top 5 people with most risk.

10. **Model Interpretation and analysis**

    a. Partial Dependence Plots

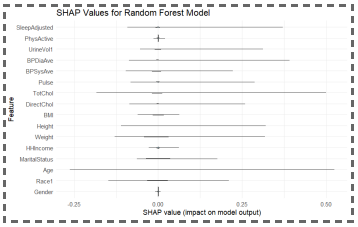
    We performed the partial dependence plots of the five most important features(Age, TotChol, Weight, BPSysAve, and Marital Status) to see how they are related with the target variable Diabetes. We used the random forest model 'best_df' to compute these plots.

    b. SHAP

    i. SHAP Values - they provide a unified measure of feature importance and allow us to understand individual predictions on Diabetes.

    ii. SHAP Summary Plot

    The plot shows how each feature contributes to the model output. Wider distributions indicate more variability in feature impact. The center of the distribution around zero suggests a neutral effect. We can see that the variables PhysActive and Gender are very centered and have very low impact on the prediction. Variables like Age or TotChol have a wider distribution and have a high effect on the prediction.



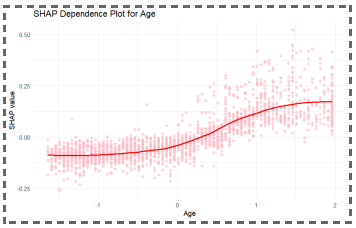
    iii. SHAP Feature Importance Plot

    Features with higher mean absolute SHAP values have a greater impact on the model's predictions. This plot provides an alternative to the traditional feature importance plot based on Gini importance.

    iv. SHAP Dependence Plots

    We performed the SHAP dependence plots of the five most important features according to the SHAP feature importance plot.

    This plot for example shows the relationship between the value of Age and its impact on the prediction. Positive SHAP values indicate an increase in the probability of having Diabetes. A trend in the plot suggests how the feature influences the model's output, which in this case shows a higher probability of having Diabetes for older people.



11. **Conclusions and best model**

Judging by the importance graphs and data frames and the partial dependency graphs we found out that the most important features in our dataset to decide whether a person has diabetes or not are Age, TotChol, Weight, Height, Race1, BPSysAve, and Marital Status. As age, weight, and BPSysAve increase, the chance of diabetes also increases, while as TotChol and Height increase, the chance of diabetes decreases.

As our dataset was very unbalanced with far less observations of people with diabetes, it was very important for our model to have good specificity as having many False Negatives is far worse than having False Positives. Hence, we found a few models that are good like the penalized logistic regression, which yielded a specificity of 88.21% and balanced_accuracy of 78.98%, and a decision tree, which had a specificity of 82.13% and balanced accuracy of 76.87%. Using the health impact with both models we get an overall positive net score, which means that the guessed true positive outweighs the false negative and false positives negative impact.